12TH CHEMISTRY · CHAPTER 22 · SOLVED EXERCISE

# Hydroxy Compounds

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

## Q1. MULTIPLE CHOICE QUESTIONS:

**I. Which compound can be oxidised to form $(CH_3)_2CHCOCH_3$?**

(a) 2-methylpropan-1-ol (b) 2,2-dimethylpropanol

(c) 2-methylbutan-2-ol (d) 3-methylbutan-2-ol

**Answer: (d) 3-methylbutan-2-ol** — The product is a ketone (3-methylbutan-2-one), so the alcohol must be the secondary alcohol with the same carbon skeleton.

**II. Which alcohol forms a mixture of alkenes when dehydrated?**

(a) propan-1-ol (b) propan-2-ol

(c) pentan-1-ol (d) pentan-2-ol

**Answer: (d) pentan-2-ol** — Hydrogen can be lost from C-1 or C-3, giving pent-1-ene and pent-2-ene.

**III. Which alcohol can be oxidised by acidified potassium dichromate(VI) but cannot be dehydrated by heating with concentrated sulfuric acid?**

(a) 2,3-dimethylbutan-2-ol (b) 2,2-dimethylpropan-1-ol

(c) 2-methylpropan-2-ol (d) pentan-3-ol

**Answer: (b) 2,2-dimethylpropan-1-ol** — It is a primary alcohol (can be oxidised), but the carbon next to the –OH carbon has no hydrogen atom, so water cannot be eliminated.

**IV. Which one of the following mechanisms is not involved in the reaction sequence below?**

**$CH_3CH_3 \rightarrow CH_3CH_2Cl \rightarrow CH_3CH_2OH \rightarrow CH_2{=}CH_2 \rightarrow CH_3CH_2Br$**

(a) electrophilic addition (b) electrophilic substitution

(c) nucleophilic substitution (d) free-radical substitution

**Answer: (b) electrophilic substitution** — The steps are free-radical substitution, nucleophilic substitution, elimination and electrophilic addition.

**V. Which one of the following does not represent an oxidation?**

(a) ethanal → ethanol (b) propan-1-ol → propanal

(c) propan-2-ol → propanone (d) ethanol → ethanoic acid

**Answer: (a) ethanal → ethanol** — An aldehyde changing to an alcohol gains hydrogen; it is a reduction.

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**VI. Dehydration of alcohol is an example of**

(a) substitution Reaction (b) addition reaction
(c) elimination Reaction (d) redox reaction

**Answer: (c) elimination Reaction** — A water molecule is removed from the alcohol to form an alkene.

**VII. Bromine reacts with phenol and decolourise orange color and turns it to which of the colored precipitate?**

(a) white precipitate (b) pink precipitate
(c) blue precipitate (d) black precipitate

**Answer: (a) white precipitate** — A white precipitate of 2,4,6-tribromophenol is formed.

**VIII. When phenol is treated with excess bromine water it gives m-bromophenol?**

(a) o- and p-bromophenol
(b) 2,4-dibromophenol
(c) 2,4,6-tribromophenol

**Answer: (c) 2,4,6-tribromophenol** — The activated ring is substituted at the 2, 4 and 6 positions.

*Only three options (a–c) are printed for MCQ VIII, and its stem wrongly ends with "it gives m-bromophenol?". The question asked is what phenol gives with excess bromine water.*

**IX. What are the correct reagents for the conversion below?**
**$CH_3CH_2CHOHCH_3 \rightarrow CH_3CH_2CHBrCH_3$**

(a) $Br_2$ and $H_2SO_4$
(b) $Br_2$ and NaOH
(c) NaBr and $H_2SO_4$
(d) NaBr and NaOH

**Answer: (c) NaBr and $H_2SO_4$** — NaBr and $H_2SO_4$ produce HBr, which replaces the –OH group by Br.

**X. Give a chemical test used in distinguishing propan-1-ol and propan-2-ol.**

(a) $Br_2$ and NaOH
(b) $SOCl_2$
(c) Acidified $K_2Cr_2O_7$
(d) $NaOH_{(aq)}$

**Answer: (a) $Br_2$ and NaOH** — Halogen with NaOH (haloform reaction) reacts only with alcohols having the $CH_3CH(OH)$– group, i.e., propan-2-ol, not propan-1-ol.

*This test is normally done with iodine and NaOH (iodoform test: yellow precipitate with propan-2-ol only); it is named in the learning outcomes but not described in the chapter. Option (c) is not suitable: both alcohols turn acidified $K_2Cr_2O_7$ from orange to green.*

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## Q2. SHORT ANSWER QUESTIONS

**a) Six alcohols are given below: i. butan-2-ol ii. ethane-1,2-diol iii. 2-methylpentan-3-ol iv. 2-methylpropan-2-ol v. propan-1-ol vi. propan-2-ol**

### i) Which alcohol is an example of a tertiary alcohol?

**iv. 2-methylpropan-2-ol**; the carbon carrying the –OH group is attached to three other carbon atoms.

### ii) Which alcohol is an example of an alcohol that can be oxidized to carboxylic acid?.

**v. propan-1-ol**; it is a primary alcohol and is oxidised to propanoic acid. (ii. ethane-1,2-diol also has primary –OH groups.)

### iii) Which alcohol is an example of an alcohol that can be oxidized to a ketone?

**vi. propan-2-ol**, a secondary alcohol, is oxidised to propanone. (i. butan-2-ol and iii. 2-methylpentan-3-ol are also secondary alcohols.)

### v) Which alcohol can't change color of acidified potassium dichromate?

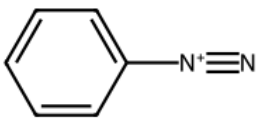
**iv. 2-methylpropan-2-ol**; tertiary alcohols are not oxidised, so the solution stays orange.

*The parts are printed as i), ii), iii), v); there is no part iv) in the textbook.*

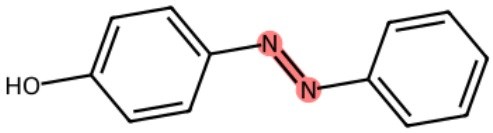
**b) Phenylamine reacts with $HNO_{2(aq)}$ at 4°C to form compound P. Compound P reacts with phenol under alkaline conditions at 4°C. The product of this reaction is acidified, forming azo compound Q.**

### i) Draw the structure of compound P and Q.

P is the benzenediazonium ion, $C_6H_5–N_2^+$. It couples with phenol at the 4-position to give Q, 4-hydroxyazobenzene, $HO–C_6H_4–N=N–C_6H_5$.




P: benzenediazonium ion




Q: 4-hydroxyazobenzene (azo group highlighted)

### ii) Circle the azo group on your structure.

The azo group is the –N=N– group joining the two benzene rings (highlighted in Q above).

### iii) State one use of an azo compound such as Q.

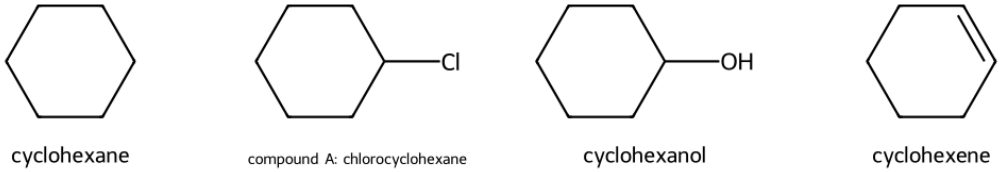
Azo compounds are used as dyes.

*The coupling of diazonium ions with phenol is not described in this chapter; the textbook defers it to Chapter 24 (Organic Nitrogen Compounds).*

**c) Cyclohexane can be converted into cyclohexene via a three stage synthesis. cyclohexane → (stage 1, $Cl_2$) compound A → (stage 2) cyclohexanol → (stage 3) cyclohexene**

**i) In stage 1, cyclohexane reacts with chlorine to form the organic product, compound A. Show the structure of compound A.**

Compound A is chlorocyclohexane, $C_6H_{11}Cl$ (formed by free-radical substitution).




cyclohexane | compound A: chlorocyclohexane | cyclohexanol | cyclohexene

**ii) Stage 3 involves the dehydration of an alcohol. State a suitable reagent for dehydrating an alcohol.**

Concentrated sulphuric acid (or concentrated phosphoric acid, or heated aluminium oxide, $Al_2O_3$).

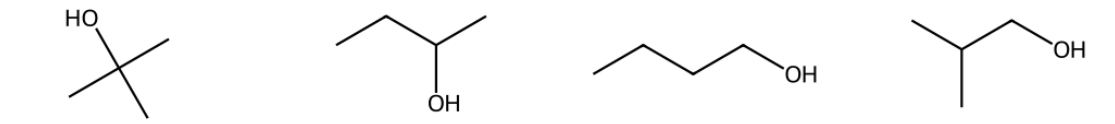
**iii) Write a balanced equation for the dehydration of cyclohexanol, $C_6H_{11}OH$.**

$$C_6H_{11}OH \rightarrow C_6H_{10} + H_2O$$

*The scheme labels the alcohol as "cyclohexannol"; the correct spelling is cyclohexanol.*

**d) There are four structurally isomeric alcohols of molecular formula $C_4H_{10}O$. Graphical formulae of these isomers, labelled A, B, C and D, are shown below.**

The four isomers are:




A: 2-methylpropan-2-ol | B (intended): butan-2-ol | C: butan-1-ol | D: 2-methylpropan-1-ol

*Structure B is printed with two $CH_3$ groups on the –OH carbon of a three-carbon chain; this is 2-methylbutan-2-ol, $C_5H_{12}O$, which is not an isomer of $C_4H_{10}O$. The fourth isomer of $C_4H_{10}O$ is butan-2-ol, which is what B should be.*

**i) Identify the type of alcohol represented by A and by B.**

A is a **tertiary** alcohol. B, as printed, is also a tertiary alcohol; the intended isomer, butan-2-ol, is a **secondary** alcohol.

**ii) Give the name of alcohol A.**

2-methylpropan-2-ol.

**iii) Select one of the alcohols A, B, C or D which will, on oxidation, produce an aldehyde. Give the structural formula of the aldehyde produced by this reaction.**

Alcohol **C** (butan-1-ol), a primary alcohol, gives butanal, $CH_3–CH_2–CH_2–CHO$. (D, 2-methylpropan-1-ol, is also primary and gives 2-methylpropanal, $(CH_3)_2CH–CHO$.)

**iv) All the alcohols A, B, C and D may be readily dehydrated. Explain what is meant by the term dehydration. State the type of compound formed by dehydration of alcohols.**

Dehydration is a type of elimination reaction in which a water molecule is removed from the alcohol. The compound formed is an **alkene**.

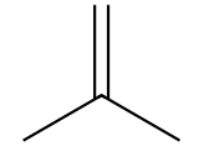
**v) Suggest suitable reagent(s) and condition(s) for the dehydration of alcohols.**

Heat the alcohol with concentrated sulphuric acid or concentrated phosphoric acid, or pass its vapour over heated aluminium oxide ($Al_2O_3$).

**e) Select one of the alcohols A, B, C or D which, on dehydration, would give a single product. Draw the structural formula of this product.**

Alcohol **A** (2-methylpropan-2-ol). All the hydrogen atoms on the neighbouring carbons are equivalent, so only one alkene, 2-methylpropene, is formed:

$$(CH_3)_3C–OH \rightarrow (CH_3)_2C=CH_2 + H_2O$$



2-methylpropene

## Q3. CONSTRUCTED RESPONSE QUESTIONS

**I. Why nitration of phenol is easier than benzene. Which product is formed when phenol is nitrated with conc. $HNO_3$ at room temperature?**

In phenol, a lone pair of the oxygen atom overlaps with the π system of the ring and becomes delocalised. This increases the electron density of the ring, so the ring is activated and is attacked by electrophiles more easily than benzene.

With concentrated $HNO_3$ the product is **2,4,6-trinitrophenol**.

**ii. Sodium hydroxide reacts with phenol but not with ethanol. Give reason.**

Phenol is a stronger acid than water because its phenoxide ion is stabilised by delocalisation of the negative charge over the ring, so it gives its proton to the hydroxide ion and forms sodium phenoxide and water. Ethanol is a weaker acid than water, because the electron-donating ethyl group destabilises the ethoxide ion, so it does not react with sodium hydroxide.

$$C_6H_5OH + NaOH \rightarrow C_6H_5O^-Na^+ + H_2O$$

### iii. Give reactions of methanol with $SOCl_2$, and $PCl_5$.

$$CH_3\text{–}OH + SOCl_2 \rightarrow CH_3\text{–}Cl + SO_2 + HCl$$
$$CH_3\text{–}OH + PCl_5 \rightarrow CH_3\text{–}Cl + POCl_3 + HCl$$

### iv. How propanol can be converted into propanal?

Propan-1-ol is a primary alcohol. It is oxidised by warming with acidified $K_2Cr_2O_7$, and the propanal is distilled off as it forms so that it is not oxidised further to propanoic acid.

$$CH_3\text{–}CH_2\text{–}CH_2\text{–}OH + [O] \rightarrow CH_3\text{–}CH_2\text{–}CHO + H_2O\ (K_2Cr_2O_7 / H^+, \text{distil})$$

### v. Write down the reaction for the substitution of each of Cl, Br, and I with –OH in 2-propanol.

The –OH group of propan-2-ol is replaced by Cl, Br and I as follows:

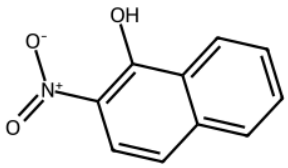
$$CH_3\text{–}CH(OH)\text{–}CH_3 + HCl \rightarrow CH_3\text{–}CHCl\text{–}CH_3 + H_2O\ (ZnCl_2)$$
$$CH_3\text{–}CH(OH)\text{–}CH_3 + KBr + H_2SO_4 \rightarrow CH_3\text{–}CHBr\text{–}CH_3 + KHSO_4 + H_2O$$
$$3CH_3\text{–}CH(OH)\text{–}CH_3 + PI_3 \rightarrow 3CH_3\text{–}CHI\text{–}CH_3 + H_3PO_3\ (\text{red P} + I_2)$$

*The iodine equation is written on the pattern of the phosphorus trihalide reaction of Chapter 21; it is not printed in this chapter.*

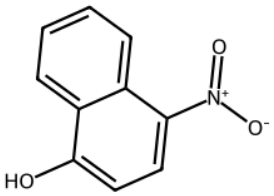
### vi. Construct a reaction for the preparation of nitro-substituted naphthol. Is there produced a single product? Explain.

1-Naphthol is treated with dilute $HNO_3$. A single product is **not** formed: the –OH group activates the ring and directs the electrophile to the 2 and 4 positions, so a mixture of 2-nitro-1-naphthol and 4-nitro-1-naphthol is obtained.

$$2C_{10}H_7OH + 2HNO_3 \rightarrow 2\text{-}O_2NC_{10}H_6OH + 4\text{-}O_2NC_{10}H_6OH + 2H_2O$$



2-nitro-1-naphthol



4-nitro-1-naphthol

## DESCRIPTIVE QUESTIONS

### Q4. Explain the acidity of phenol

Although phenols contain an –OH group, they are weakly acidic. This is due to the delocalisation of one of the lone pairs of the oxygen atom into the aromatic ring: the lone pair overlaps with the delocalised π system of the ring. The delocalisation stabilises the **phenoxide ion**, the conjugate base of phenol: the negative charge is spread over the aromatic ring, so there is less charge density on the oxygen atom. The $H^+$ ions are therefore not strongly attracted to the phenoxide ion and are less likely to re-form the phenol molecule. So phenol is more likely to lose a proton (act as an acid) than to gain one.

$$C_6H_5OH_{(aq)} \rightleftharpoons C_6H_5O^-_{(aq)} + H^+_{(aq)}$$

Phenol is therefore a stronger acid than water and alcohols, and it reacts with bases such as NaOH and with sodium metal.

Relative acidities: phenol > water > ethanol

### Q5. Explain why the reagents and conditions for the nitration and bromination of phenol are different from those for benzene.

In phenol the oxygen atom of the hydroxyl group donates electron density into the ring: one of its lone pairs overlaps with the π system of the benzene ring and becomes delocalised. This increases the electron density in the aromatic ring, so the ring is more likely to undergo electrophilic attack; it is **activated**. The incoming electrophiles are directed by the hydroxyl group to the 2, 4 and 6 positions.

Because the ring of phenol is activated, milder reagents and conditions are enough, and more than one hydrogen atom can be substituted:

| Reaction | Benzene | Phenol |
|---|---|---|
| Nitration | Mixture of conc. $HNO_3$ and conc. $H_2SO_4$, refluxed between 25 °C and 60 °C; gives nitrobenzene. | Dilute $HNO_3$ at room temperature; gives 2- and 4-nitrophenol. Conc. $HNO_3$ gives 2,4,6-trinitrophenol. |
| Bromination | Pure bromine (not a solution) and anhydrous $AlBr_3$ catalyst at room temperature; gives bromobenzene. | Bromine water at room temperature, no catalyst; gives a white precipitate of 2,4,6-tribromophenol. |

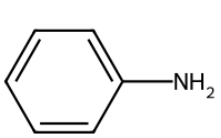
### Q6. Describe how phenylamine can be converted into phenol, including reagents and conditions.

Phenols can be produced by the reaction of phenylamine with nitrous acid ($HNO_2$). The nitrous acid is produced *in situ*. The reaction occurs in three steps:

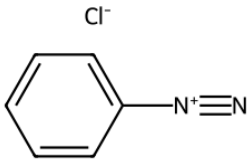
1. $HNO_2$ is so unstable that it is prepared in the test-tube by reacting sodium nitrite ($NaNO_2$) with dilute hydrochloric acid (HCl), keeping the temperature below 10 °C using ice.

2. Phenylamine reacts with the $HNO_2$ to form an unstable diazonium salt.
3. On further warming, the diazonium salt is decomposed (hydrolysed) and phenol is produced.

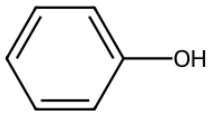
$$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$$
$$C_6H_5NH_2 + HNO_2 + HCl \rightarrow C_6H_5N_2^+Cl^- + 2H_2O \text{ (below 10 °C)}$$
$$C_6H_5N_2^+Cl^- + H_2O \rightarrow C_6H_5OH + N_2 + HCl \text{ (warm)}$$



phenylamine



benzenediazonium chloride



phenol

*The textbook shows the sequence as a scheme ($NaNO_2$, HCl, < 10 °C; then warming, with NaOH written on the second arrow and NaCl + $N_2$ as by-products). The balanced equations above are the standard ones; the learning outcome says "warming of the diazonium salt with $H_2O$".*

## Q7. Explain why ethanol is less acidic than water, but water is less acidic than phenol.

**Ethanol and water:** both dissociate only slightly, but ethanol gives a lower $[H^+]$ than water. The ethyl group is electron-donating (positive inductive effect); it pushes electrons towards the oxygen of the ethoxide ion, increases its electron density and destabilises it, so the ion readily accepts $H^+$ to form ethanol again. The hydroxide ion has no such group. So ethanol is less acidic than water.

**Water and phenol:** in the phenoxide ion the negative charge is delocalised over the benzene ring, so the charge density on oxygen is lower than in the hydroxide ion. The phenoxide ion is more stable and attracts $H^+$ less strongly, so phenol dissociates more than water.

| | Ethanol | Water | Phenol |
|---|---|---|---|
| Conjugate base | ethoxide ion, $C_2H_5O^-$ | hydroxide ion, $OH^-$ | phenoxide ion, $C_6H_5O^-$ |
| Effect on the negative charge | Ethyl group donates electrons (+I): more electron density on O; ion destabilized | No such effect | Charge delocalised over the benzene ring: ion stabilized |
| Acid strength | Weakest | Intermediate | Strongest of the three |

Relative acidities: phenol > water > ethanol

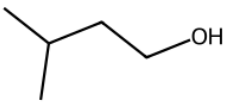
## QUICK CHECK QUESTIONS (IN-CHAPTER)

*These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.*

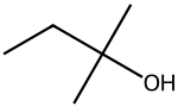
### Quick Check 22.1

**I. Which ones of these are 1°, 2° and 3° alcohols?**

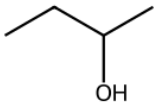
| | Alcohol | –OH carbon is attached to | Type |
|---|---|---|---|
| a) | 3-methylbutan-1-ol | one carbon atom | 1° |
| b) | 2-methylbutan-2-ol | three carbon atoms | 3° |
| c) | butan-2-ol | two carbon atoms | 2° |
| d) | pentan-1-ol | one carbon atom | 1° |
| e) | 3-methylbutan-2-ol | two carbon atoms | 2° |
| f) | cyclopentanol | two ring carbon atoms | 2° |



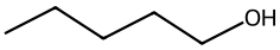

a) 3-methylbutan-1-ol



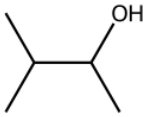

b) 2-methylbutan-2-ol



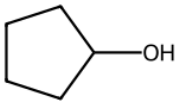

c) butan-2-ol




d) pentan-1-ol




e) 3-methylbutan-2-ol




f) cyclopentanol

### Quick Check 22.2

**i. Compare the relative acidic strength of methanol and ethanol.**

Methanol is slightly more acidic than ethanol. Alkyl groups donate electrons (positive inductive effect) and destabilize the alkoxide ion. The ethyl group donates more electron density than the methyl group, so the ethoxide ion is less stable than the methoxide ion and ethanol dissociates less.

**ii. Showing the electron delocalisation explain why the phenoxide ($PhO^-$) is more stable than the ethoxide ion ($C_2H_5O^-$).**

In the phenoxide ion, a lone pair on the oxygen overlaps with the delocalised π system of the benzene ring. The negative charge is spread over the oxygen and the ring (extended π system), so the charge density on oxygen is low and the ion is stable. In the ethoxide ion there is no π system; the negative charge stays on the oxygen and is even increased by the electron-donating ethyl group, so the ion is less stable.

**iii. Alkali metals react with stronger acids to produce the hydrogen gas. A piece of sodium is added to two beakers; one containing ethanol and the other phenol. a) Predict in which beaker the hydrogen gas will be evolved earlier and more rapidly. b) If the phenolphthalein indicator is already present before adding sodium to the above solutions, which solution will turn pink earlier?**

**a)** In the beaker containing **phenol**. Phenol is a stronger acid than ethanol, so it gives up its proton to sodium more readily and hydrogen is evolved earlier and more rapidly.

**b)** The pink colour appears when a base is formed. The **ethanol** solution is expected to turn pink earlier: the ethoxide ion formed is a strong base. In the phenol beaker the phenoxide ion formed is a much weaker base (it is stabilised by delocalisation), and the remaining phenol is itself acidic, so the solution becomes alkaline enough for the indicator only later.

*Part b) is answered by reasoning from the relative base strengths of ethoxide and phenoxide; the textbook only states that sodium with ethanol and phenolphthalein gives the pink colour slowly, and with water immediately.*

## Quick Check 22.3

**a) Write chemical reaction of propan-1-ol with $SOCl_2$.**

$$CH_3–CH_2–CH_2–OH + SOCl_2 \rightarrow CH_3–CH_2–CH_2–Cl + SO_2 + HCl$$

The product is 1-chloropropane; both by-products are gases.

**b) Differentiate the reaction of $PCl_3$ and $PCl_5$ with propan-1-ol with the help of the chemical equations.**

$$3CH_3–CH_2–CH_2–OH + PCl_3 \rightarrow 3CH_3–CH_2–CH_2–Cl + H_3PO_3$$
$$CH_3–CH_2–CH_2–OH + PCl_5 \rightarrow CH_3–CH_2–CH_2–Cl + POCl_3 + HCl$$

With $PCl_3$, three moles of alcohol react with one mole of the reagent and phosphorous acid ($H_3PO_3$) is the by-product. With $PCl_5$, one mole of alcohol reacts with one mole of the reagent; the by-products are $POCl_3$ and steamy fumes of HCl.

**c) How ethanol reacts with KBr? give reaction equation.**

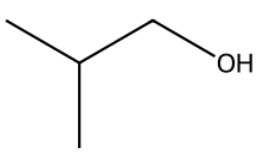
Ethanol is heated with potassium bromide and concentrated sulphuric acid. The acid first reacts with KBr to give HBr, which then reacts with ethanol to give bromoethane.

$$CH_3–CH_2–OH + KBr + H_2SO_4 \rightarrow CH_3–CH_2–Br + KHSO_4 + H_2O$$

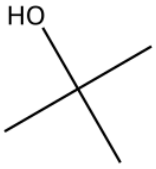
## Quick Check 22.4

**a) Three alcohols A, B and C are structural isomers of $C_4H_{10}O$. Each alcohol is refluxed with acidified dichromate(VI), $H^+/Cr_2O_7^{2-}$. i) Write the systematic name for alcohol C. ii) Write the equation for the complete combustion of alcohol A.**

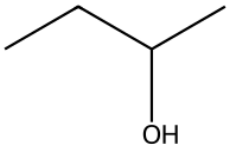
The three structures are shown below (i = A, ii = B, iii = C).




A: 2-methylpropan-1-ol




B: 2-methylpropan-2-ol




C: butan-2-ol

**i)** Alcohol C is **butan-2-ol**.

**ii)** Alcohol A is 2-methylpropan-1-ol:

$$C_4H_{10}O + 6O_2 \rightarrow 4CO_2 + 5H_2O$$

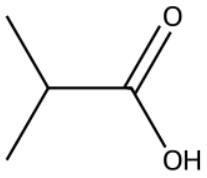
**b) Acidified potassium manganate(VII), $KMnO_4$, is a purple oxidising agent. What colour change would you observe when alcohols are oxidised with manganate ions ($MnO_4^-$)?**

The purple colour of the $MnO_4^-$ ions disappears: the solution turns from **purple to colourless**, as $MnO_4^-$ is reduced to nearly colourless $Mn^{2+}$ ions. (This happens with primary and secondary alcohols only.)

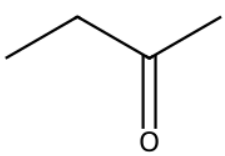
*The ion is printed as "$MnO^{4-}$" in the textbook; the manganate(VII) ion is $MnO_4^-$.*

**c) Draw the structures for the organic products of the oxidation of the following. If there is no reaction, write 'NONE'.**

| Alcohol | Type | Organic product (reflux with acidified dichromate) |
|---|---|---|
| A: 2-methylpropan-1-ol | Primary | 2-methylpropanoic acid, $(CH_3)_2CH–COOH$ (via 2-methylpropanal) |
| B: 2-methylpropan-2-ol | Tertiary | NONE |
| C: butan-2-ol | Secondary | butanone, $CH_3–CO–CH_2–CH_3$ |




from A: 2-methylpropanoic acid




from C: butanone

*No separate structures are printed for part c); it is answered for alcohols A, B and C of part a).*

## Quick Check 22.5

**a) The reaction of phenol with bromine water is used for the identification of phenol. What will be the visual change during the reaction, also give the systematic name of the compound formed.**

The orange colour of bromine water is decolourised and a **white precipitate** is formed. The compound is **2,4,6-tribromophenol**.

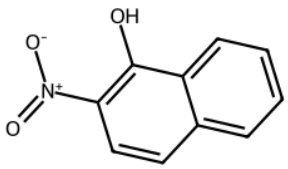
**b) Nitration is an example of aromatic electrophilic substitution and its rate depends upon the group already present in the benzene ring. Out of benzene and phenol, which one is more easily nitrated and why?**

**Phenol.** A lone pair on the oxygen of the –OH group is delocalised into the ring, which increases the electron density of the ring and activates it towards electrophiles. So phenol is nitrated by dilute $HNO_3$ at room temperature, whereas benzene needs a mixture of conc. $HNO_3$ and conc. $H_2SO_4$.

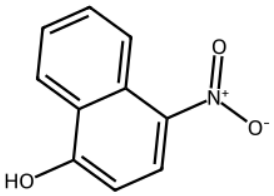
**c) Write the reaction equation for nitration of 1-naphthol.**

The –OH group directs the nitro group to the 2 and 4 positions, so a mixture of 2-nitro-1-naphthol and 4-nitro-1-naphthol is formed (dilute $HNO_3$, room temperature):

$$2C_{10}H_7OH + 2HNO_3 \rightarrow 2\text{-}O_2NC_{10}H_6OH + 4\text{-}O_2NC_{10}H_6OH + 2H_2O$$




2-nitro-1-naphthol




4-nitro-1-naphthol